be a little larger, augmentied by the mass of everything that has fallen into the hole since it was born: interstellar gas, rocks, starships . . .

You know all this because, before embarking on your voyage, you studied the fundamental laws of gravity: laws that were discovered in an approximate form by Isaac Newton in 1687, and were radically revised into a more accurate form by Albert Einstein in 1915.[9] You learned that Einstein's gravitational laws, which are called *general relativity,* force black holes to behave in these ways as inexorably as they force a dropped stone to fall to earth. It is impossible for the stone to violate the laws of gravity and fall upward or hover in the air, and similarly it is impossible for a black hole to evade the gravitational laws: The hole must be born when a star implodes upon itself; the hole's mass, at birth, must be the same as the star's; and each time something falls into the hole, its mass must grow[10]. Similarly, if the star is spinning as it implodes, then the newborn hole must also spin; and the hole's *angular momentum* (a precise measure of how fast it spins) must be the same as the star's.

Before your voyage, you also studied the history of human understanding about black holes. Back in the 1970s Brandon Carter, Stephen Hawking, Werner Israel, and others, using Einstein's general relativistic description[11] of the laws of gravity, deduced that a black hole must be an exceedingly simple beast[12]: All of the hole's properties—the strength of its gravitational pull, the amount by which it deflects the trajectories of starlight, the shape and size of its surface—are determined by just three numbers: the hole's mass, which you now know; the angular momentum of its spin, which you don't yet know; and its electrical charge. You are aware, moreover, that no hole in interstellar space can contain much electrical charge; if it did, it quickly would pull opposite charges from the interstellar gas into itself, thereby neutralizing its own charge.

As it spins, the hole should drag the space near itself into a swirling, tornado-like motion relative to space far away, much as a spinning airplane propeller drags air near itself into motion; and the swirl of space should cause a swirl in the motion of anything near the hole.[13]

To learn the angular momentum of Hades, you therefore look for a tornado-like swirl in the stream of interstellar gas atoms as they fall into the hole. To your surprise, as they fall closer and closer to the hole, moving